
An Analysis of Inductive Biases and Generalization of Deep Learning Models (ATML-PA1)

Abstract

We study the inductive biases of discriminative, generative, and contrastive vision models and their effect on generalization. For discriminative models, CNNs and ViTs are fine-tuned and evaluated on STL-10 and its variations to assess reliance on texture, shape, and color, revealing strong texture bias in CNNs and shape/semantic bias in ViTs. For generative models, VAEs and GANs are trained on CIFAR-10 and tested on out-of-distribution data to examine sample diversity, fidelity, and mode coverage, highlighting VAEs' mode-covering bias and GANs' mode-seeking bias. For contrastive multimodal models, CLIP is evaluated zero-shot to probe shape and semantic reliance. Results show that architecture and large-scale, diverse training induce human-aligned biases, supporting robust OOD generalization and resilience to domain shifts. Code and reproducibility material is available at: [code](#).

1. Introduction

Deep learning models have inherent inductive biases that affect how they respond to and treat different inputs and consequently, their performance and generalization. It is crucial that these biases are understood and evaluated to understand model failures and guide future model design. Inductive biases can be categorized into three main types: i) semantic biases ii) architectural biases iii) property-wise biases.

Inductive biases in vision models can be broadly categorized into three types: (i) *semantic biases*, such as reliance on visual cues like shape or texture; (ii) *architectural biases*, which encode properties such as translation invariance or permutation sensitivity; and (iii) *property-wise biases*, arising from underlying operations (e.g., convolutions emphasizing locality, or attention enabling global context).

In this work, we study these biases in discriminative, generative and contrastive models. For discriminative models, we compare pretrained ResNet-50 (CNN) (He et al., 2015) and ViT-B/16 (vision transformer) (Dosovitskiy et al., 2021). By testing on carefully constructed dataset variants, we probe (i) semantic biases (shape, texture, color sensitivity), (ii)

architectural biases (translation and permutation invariance), and (iii) property-wise biases through out-of-distribution generalization.

For generative models, we examine VAEs and GANs. VAEs exhibit a mode-covering bias due to the forward KL in their objective, leading to diverse but blurrier reconstructions, whereas GANs, by optimising a mode-seeking divergence, produce sharper samples at the cost of partial mode collapse.

For contrastive multimodal models, we study CLIP (Radford et al., 2021) to understand how large-scale vision-language pretraining shapes inductive biases and generalization. We evaluate zero-shot performance on CIFAR-10, PACS, and stylized images (sketches, cartoons, paintings), examine the impact of textual prompt design, and probe reliance on shape versus fine-grained texture using texture perturbations. The following sections detail the experimental setup and evaluation framework.

Through these experiments, we aim to answer the following questions:

- [RQ1] Do CNNs and ViTs rely on different visual cues when recognizing images?
- [RQ2] How do built-in architectural assumptions affect model properties like translation invariance and permutation sensitivity?
- [RQ3] How do different inductive biases impact generalization to out-of-distribution data?
- [RQ4] In what ways do a VAE and a GAN differ in the representations they learn and the images they generate?
- [RQ5] What biases emerge in a contrastively trained vision-language model like CLIP compared to a standard vision model?

2. Methodology

2.1. Discriminative Models (CNNs and ViTs)

To evaluate inductive biases and domain generalization in discriminative models, we use two pretrained models: ResNet-50 as a representative CNN and ViT-B/16 as a representative ViT - both SOTA models for their respective architectures. These choices ensure fairness and robust baselines. Since both models are trained on the extremely



Figure 1. Example images from the stylized STL-10 dataset.

large ImageNet dataset, the effects of other confounding factors like training variation would be limited, ensuring a reliable study.

2.1.1. EVALUATING INDUCTIVE BIASES

We use the STL-10 dataset (Coates et al., 2011) as the baseline to evaluate inductive biases, owing to its reasonable size and better visualizations, as a result of its higher resolution images, over other datasets like CIFAR-10. The dataset contains 13,000 (5000 train, 8000 test) 96×96 colored images, spanning 10 classes. For both ResNet-50 and ViT-B/16, the backbone is frozen and the classification head is replaced with a linear head outputting 10 classes. To prepare the data, all images are resized to 224×224 pixels and normalized using statistics. Both models are fine-tuned on the training set following the same training regime - for 5 epochs, using the cross-entropy loss and the Adam optimizer (Kingma & Ba, 2017) with a fixed learning rate of 0.005 and batch size of 64. These fine-tuned models are used for all further evaluations.

We use accuracy as our baseline metric and use any additional metrics necessary. The models are first evaluated on the clean (original) test set to ensure that the models perform well on in-distribution data. We then evaluate these models on carefully curated modified versions of the STL-10 dataset where each modification is used to highlight a models inductive bias. The following modifications are made:

Grayscale: test images are converted to grayscale, while keeping the number of channels 3 (expected model input), to evaluate the color bias. We measure the drop in accuracy compared to the clean set to quantify color sensitivity.

Stylized Cue conflict: Using adaptive instance normalization (AdaIN) (Huang & Belongie, 2017), the test set is stylized with artistic styles, removing any texture information, to generate a cue conflict dataset to study the shape vs. texture bias of these models (Fig. 1). We calculate the shape bias for both models as the fraction of accuracy on the stylized dataset over the clean dataset to understand their reliance on texture/shapes.

Shifted: test images are translated by a few pixels in all directions to assess translation invariance. The images are

shifted by 10 pixels in all directions (up, down, left, right, up left, up right, down left, down right). We measure the consistency i.e the fraction of images get the same top-1 prediction after shifting.

Patch shuffled: test images are divided into patches and then shuffled to disrupt spatial structure to evaluate permutation invariance. A patch size of 32 is used to create a 7×7 grid, which are then shuffled.

Patch occluded: patches are randomly masked to disrupt global coherence to evaluate the reliance on global context vs. local features. A patch size of 16 is used to create a 14×14 grid, from which 12 patches are masked with black patches.

Finally, to understand the different between how the model represents the data, we visualize the feature space through t-SNE, using the features from the penultimate layers of both models. We perform this for the clean and stylized dataset to understand the model’s feature representation. Together, this setup performs a comprehensive evaluation of the inductive biases present in discriminative models.

2.1.2. EVALUATING DOMAIN GENERALIZATION

We use the PACS dataset (Li et al., 2017) to evaluate the OOD generalization of the models. The dataset is a classical example of covariate shift, consisting of four domains (photo, art, cartoon, and sketch) in the same label space. Each domain contains 2500 images across 7 object classes with an average of 357 images per class.

We train both models on the photo, art, and cartoon domains and leave the sketch domain out for evaluation, owing to its challenging nature. For training, we adopt the same training strategy as the one used for the STL-10 dataset. Once trained, we calculate the accuracy on the sketch domain for both models to evaluate their OOD performance.

2.2. Generative Models (VAE vs. GAN)

2.2.1. DATA AND PREPROCESSING

We used CIFAR-10, CIFAR-100, and SVHN, all consisting of natural images of size 32×32 . CIFAR-10 provides 50,000 training and 10,000 test images, with the training set split into 45,000 train and 5,000 validation. For CIFAR-100 and SVHN, only the test splits were used (10,000 and $\sim 23,000$ images, respectively). All datasets were normalized to the range $[-1, 1]$.

2.2.2. ARCHITECTURES AND TRAINING

Our VAE uses an encoder with three downsampling blocks with batch normalization and ReLU, mapping inputs from 32×32 to 4×4 before a 128-dimensional latent. The decoder mirrors this structure with three upsampling blocks

(linear interpolation and convolution to avoid checkerboard artifacts, followed by a second convolution for expressivity) and a final $\tanh$ output. The GAN follows a DCGAN-style design, the generator maps a 128-D Gaussian latent to 32×32 images using a progressive interpolation and convolution blocks with batch norm and ReLU, ending with $\tanh$. The discriminator mirrors the VAE encoder but replaces batch norm with spectral normalization to stabilize training. The VAE was trained for 50 epochs with Adam ($\text{lr} = 10^{-4}$), using KL annealing where β increases linearly from 0 to 1 over 20k steps (14 epochs). The GAN was trained with the non-saturating DCGAN loss for 50 epochs using Adam ($\text{lr} = 2 \times 10^{-4}$, $\beta_1 = 0.5$, $\beta_2 = 0.999$).

2.2.3. OUTPUT QUALITY AND DIVERSITY

In order to judge the visual quality of the images produced by the VAE, we reconstructed 5 randomly selected images from the CIFAR-10 test set and visualised them. For GAN, which generates images unconditionally from latent noise, we sampled 5 random vectors from a standard normal distribution and passed them through the generator for visualization. In order to test whether our GAN models the entire distribution or just a specific mode of the distribution we fine-tuned an ImageNet-pretrained ResNet-152 on CIFAR-10. We then used this classifier to label 10,000 images generated by the GAN. Only predictions with confidence above a chosen threshold were retained, and we measured the number of CIFAR-10 classes represented among these high-confidence samples. This analysis allowed us to quantify mode collapse.

Moreover, we wanted to evaluate which model more closely represented the data distribution. For this task, we turned to the Fréchet Inception Distance (FID), a standard metric in generative modeling for quantifying distributional similarity, allowing us to benchmark our models against the ground-truth distribution. To compute the FID, we used an Inception-V3 network pretrained on ImageNet. For the VAE, we reconstructed CIFAR-10 test images and computed the corresponding feature means and covariances. For the GAN, we sampled 10,000 latent vectors from a standard normal distribution, generated images, and extracted the same statistics. The ground-truth CIFAR-10 test images were used as the reference distribution and then FID was calculated.

2.2.4. LATENT REPRESENTATION ANALYSIS

For both models, we wanted to visualize whether latent interpolations produce smooth transitions. For the VAE, we encoded two CIFAR-10 test images into latent vectors z_1 and z_2 and linearly interpolated between them to generate reconstructions. For the GAN, we sampled two latent vectors from $\mathcal{N}(0, I)$ and interpolated between them to produce

intermediate samples.

VAEs usually do not cluster the embedding space by classes. In order to test this hypothesis we got mean vectors against the entire test set of CIFAR-10. We employed t-SNE to reduce the dimensions to 2 for visualization.

To study the role of individual latent dimensions, we used a traversal based strategy. Given an input x , the VAE encoder provides $(\mu, \log \sigma^2)$ for the posterior $q(z|x)$. We fix all latent dimensions at μ and sweep one dimension z_d across a linear grid $\mu_d \pm k\sigma_d$ with $k = 3$ in $s = 7$ steps. For each sweep, the decoder output is compared to the baseline reconstruction at μ using mean squared error (MSE). The average MSE across sweeps defines the *effect size* dimension d , quantifying its visual influence. We rank dimensions by effect size, visualize the most influential ones through traversal grids, to highlight which latent coordinates carry the most salient generative factors.

2.2.5. OUT-OF-DISTRIBUTION GENERALIZATION

We evaluated the trained VAE on the test sets of CIFAR-100 and SVHN to examine its behavior on unseen datasets. We measured reconstruction MSE and KL divergence and visualized reconstructions to check how well the model generalizes beyond CIFAR-10.

For the GAN, OOD inputs represent latent samples drawn far from the standard normal prior $\mathcal{N}(0, I)$. To simulate this, we sampled from $\mathcal{N}(0, 10^2 I)$ and visualized the generated images to examine how the model behaves when latent codes deviate strongly from its training distribution.

In order to evaluate whether the VAE could serve as an anomaly detector, we measured the difference in error metrics it achieved on CIFAR-10 and SVHN. For each image, we computed the ELBO and to decide decision thresholds we use one half of CIFAR-10 to estimate the lower and upper quantiles of the reconstruction error distribution, which served as cutoffs for normal vs. anomalous scores, and the other half was used to estimate the false positive rate on CIFAR-10. The SVHN test set was then passed through the VAE, and its reconstruction errors were compared against these thresholds to obtain detection rates. Furthermore, we reported AUROC and AUPRC. By choosing different percentile cutoffs used for thresholding, we observed how the trade off between false positives and detection rate changed.

2.3. Contrastive Models (CLIP)

2.3.1. ZERO-SHOT CLASSIFICATION ON CIFAR-10

We began by using a pretrained CLIP model from Hugging Face, specifically the ViT-B/32 model along with its corresponding preprocessor. For evaluation, we first employed the CIFAR-10 dataset, and manually created a list

of text prompts for each of the CIFAR classes to be used in zero-shot classification. Each text prompt was defined as “a photo of a {cifar class}” to be as precise as possible while still maintaining consistency. Once tokenized, the text prompts were passed through the CLIP text encoder to produce dense vector embeddings, which were then normalized to unit length using L2 normalization.

Next, we prepared to evaluate the model on the CIFAR-10 test set. As with the text embeddings, image embeddings were normalized to unit length so that both modalities resided in the same embedding space. For each batch of images, we computed cosine similarity between image embeddings and text embeddings corresponding to all classes. The predicted label was chosen as the class with the maximum similarity score. We tracked both overall accuracy and per-class accuracy, reporting the final zero-shot accuracy as well as class-wise breakdowns to highlight categories where CLIP performance.

2.3.2. DOMAIN SHIFT EVALUATION WITH PACS

In order to evaluate CLIP under domain shift, we then used the PACS dataset which provided a natural test of CLIP’s ability to generalize across domains, since the underlying class (e.g., “dog”, “horse”) is preserved but the visual appearance changes dramatically. To maximize zero-shot performance, we constructed prompt ensembles rather than relying on a single wording. For each class, we created five different text prompts. During evaluation, we averaged across prompt variants to form robust class representations, enabling CLIP to vote across different textual descriptions.

2.3.3. PROMPT ENGINEERING

To investigate CLIP’s zero-shot performance, we experimented with prompt engineering to examine how the wording of text prompts affects classification accuracy. As highlighted in the original CLIP paper, language priors play a crucial role in guiding the model, and slight changes in phrasing can shift performance significantly. We defined four prompt sets, simple, photo, detailed and ensemble. We then ran evaluations across all four prompt sets to analyze the effect of wording on performance.

2.3.4. IMAGE-TEXT RETRIEVAL EXPERIMENT

We tested CLIP’s multimodal alignment capabilities with a small image–text retrieval experiment. We sampled 20 random images from the sketch domain and recorded their ground-truth labels. These images were preprocessed and passed through CLIP’s image encoder to obtain embeddings. Similarly, a set of text queries (e.g., “a sketch of a dog”, “a sketch of a guitar”, “a sketch of a person”) were encoded into text embeddings. Both image and text embeddings were normalized, and two retrieval tasks were performed:

Table 1. Accuracy comparison of fine-tuned ResNet-50 and ViT-B/16 across variations of the STL-10 dataset.

DATA MODIFICATION	RESNET-50 (%)	ViT-B/16 (%)
CLEAN	96.72	98.51
GRAYSCALE	92.17	96.22
STYLIZED	41.39	77.50
SHIFTED	96.04	97.15
PATCH SHUFFLED	74.15	83.95
PATCH OCCLUDED	94.58	98.29

(1) Text to Image and (2) Image to Text Retrieval.

2.3.5. EMBEDDING VISUALIZATION

In order to analyze CLIP’s representations more deeply, we set up an embedding visualization experiment comparing CLIP with a supervised baseline. We constructed a mixed dataset combining 50 CIFAR-10 test images with 30 samples each from three PACS domains (sketch, cartoon, and photo), as well as an additional grayscale-OOD subset. Each image was passed through CLIP’s image encoder to extract normalized embeddings, which were then reduced to 2D with t-SNE. We visualized embeddings colored both by domain and by semantic class to assess whether CLIP grouped images primarily by semantics or by surface-level domain characteristics.

2.3.6. SHAPE-TEXTURE BIAS ANALYSIS

We explicitly tested CLIP’s shape–texture bias using a cue conflict dataset. The dataset contained stylized images where the shape belonged to one class while the texture was borrowed from another. We loaded images from the cat folder containing such shape–texture conflicts. For each image, we computed cosine similarity scores between the CLIP image embedding and all class text embeddings (constructed with prompts like “a photo of a {class}”). This setup allowed us to examine whether CLIP’s predictions aligned more with global shape or local texture cues under conflicting conditions.

2.3.7. ROBUSTNESS UNDER IMAGE PERTURBATIONS

We evaluated CLIP’s robustness by testing a single cat image under various perturbations, including Gaussian blur, grayscale, edge detection, solarization, and posterization, simulating ImageNet-C and ImageNet-R conditions. Each variant was encoded and compared against normalized text embeddings for three prompts, allowing us to qualitatively assess CLIP’s resilience to distortions and stylistic changes.

3. Results

3.1. Reliance on Visual Cues in CNNs and ViTs

The fine-tuned ResNet-50 and ViT-B/16 achieved 96.72%, and 98.51% accuracy on the test set of STL-10. The high classification accuracy showed the models were able to learn robust interpretations for the in distribution data, regardless of the model choice. The ViT outperformed the CNN, which can be associated with its inductive biases. The reliance on visual cues and the semantic bias found in these models were evaluated and the accuracy for the models on all versions of the STL-10 dataset are shown in Table 1.

3.1.1. COLOR BIAS

ResNet-50 achieved an accuracy of 92.17% on the grayscale dataset whereas the ViT-B/16 model achieved an accuracy of 96.22%. While the CNN model was still able to achieve a relatively high accuracy, it suffered a significant accuracy drop of 4.55% from the baseline. On the contrary, the ViT achieved a higher accuracy with only a drop of 2.29% in accuracy. This can be attributed to the model learning global features like structure and shape.

3.1.2. SHAPE VS. TEXTURE BIAS

The stylized dataset removed any texture information that was previously present in the clean test set. The ResNet failed to classify images and suffered significantly, only achieving an accuracy of 41.39% on the cue-conflict dataset. The ViT also suffered an accuracy drop but still achieved an accuracy of 77.50% highlighting its robustness to changes in texture and reliance on shape, since it is preserved in the dataset. This was quantified using the shape bias, with the ResNet having a shape bias of 42.79% and the ViT having a shape bias of 78.66%, once again highlighting its reliance on shape (Fig. 2).

Using the clean and stylized dataset, the embeddings were extracted from the penultimate layers and plotted using t-SNE. The plots are shown in Figure 5. The ViT shows far better class separation with the clusters having a smaller spread, highlighting stronger discriminative ability. The ResNet’s feature, despite showing substantial class separation, are far more spread out. Both models are able to distinguish between classes for the normal dataset to a large extent with the ViT still outperforming the CNN. However, CNN struggles significantly more on the stylized dataset with samples from all classes clustering around the center. While this is also true for the ViT, its inter-class margins are more separated and the number of misplaced samples are far less in frequency. This signifies that the ViT is better able learn transferable representations.

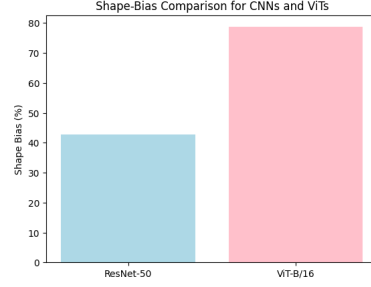


Figure 2. Shape-bias comparison of ResNet-50 and ViT-B/determined through stylized STL-10.

3.2. Effect of Architectural Biases on Invariance

3.2.1. TRANSLATION INVARIANCE

ResNet-50 and ViT-B/16 achieved accuracies of 96.04% and 97.85% respectively on the shifted dataset. Both remained robust with the CNN showing higher translation invariance owing to the negligible differences between the baseline accuracy. The CNN achieved a higher consistency of 0.98 whereas the ViT achieved a lower consistency of 0.94, highlighting the CNN’s translation invariance. The ViT showed a larger deviation from the baseline but still did not fail under the conditions of the shifted dataset.

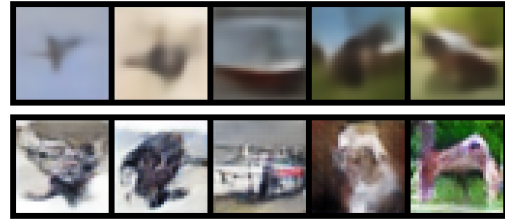


Figure 3. Reconstructions from the VAE (top row) and samples generated by the GAN (bottom row).

3.2.2. PERMUTATION SENSITIVITY

For the patch shuffled dataset, ViT-B/16 achieved an accuracy of 83.95% whereas ResNet-50 achieved an accuracy of 74.15%. The ViT showed greater robustness to the shuffling of the images and lower permutation sensitivity. For the patch occluded dataset, a similar pattern was observed with the ViT remaining largely unaffected and achieving an accuracy of 98.29%. ResNet-50, although suffering from a larger accuracy drop of 2.14% still showed robustness. Overall, ViT-B/16 consistently exhibited lower sensitivity to spatial permutations and occlusions than ResNet-50.

3.3. OOD evaluation

On the in distribution data, ResNet-50 and ViT-B/16 achieved validation accuracies of 82.04% and 94.40% respectively, highlighting the ViT’s ability to better handle

more complex data. However, both models failed in the domain shift, when evaluated on the sketch domain. ResNet-50 achieved an accuracy of 53.14% and the ViT achieved 46.12%. ResNet failed due to its inherent texture bias since the sketch images have no texture and contain only outlines/edges. The ViT also failed and achieved an accuracy even lower than the CNN, which could be associated with the intrinsic difficulty of the sketch domain.

3.4. Generative Models (VAE vs GAN)

3.4.1. OUTPUT QUALITY AND DIVERSITY

The reconstructions from the VAE and the samples generated from the GAN have been shown in Figure 3. The VAE reconstructions preserve overall class structure but are often blurrier compared to the GAN outputs. In contrast, GAN samples have sharper textures but sometimes contain unrealistic artifacts/noise.

The class distribution of 10,000 GAN generated samples, evaluated with our fine-tuned ResNet-152 at a confidence threshold 60%, is shown in Figure 4. All CIFAR-10 classes are represented, indicating that the generator does not suffer from strict mode collapse. However, the distribution is imbalanced, suggesting that there is some partial mode collapse

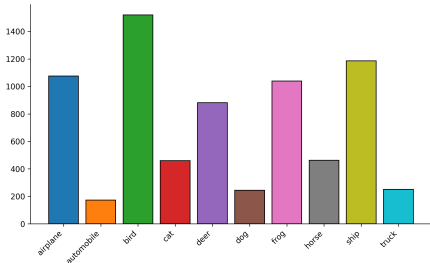


Figure 4. Class distribution of 10,000 GAN-generated samples evaluated with a fine-tuned ResNet-152 at a 60% confidence threshold.

Finally, we used FID to quantify the distance between the distribution of the generated images and that of the real dataset. The FID distance of our VAE to the original distribution is 132 while the FID distance of our GAN to the original distribution comes around 30 which tells us that the samples generated by our GAN is much closer to the original distribution comes to be 30.

3.4.2. LATENT SPACE BEHAVIOR

In the case for latent space visualization, the generated images transition smoothly as the latent representation moves from z_1 to z_2 .

Figure 7 shows the latent mean vectors of the CIFAR-10 test

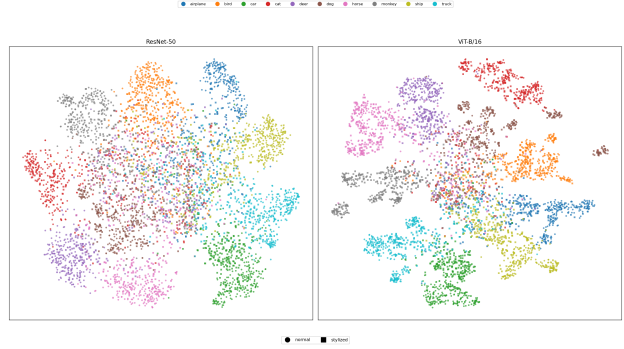


Figure 5. t-SNE of ResNet-50 and ViT-B/16 embeddings on clean vs. stylized STL-10.

set, projected into two dimensions using t-SNE. Each point is color-coded by its class label. The VAE latent means do not form clear class-specific clusters. While small pockets of same-class points can be observed, the overall distribution exhibits heavy overlap across classes, indicating that the VAE does not separate the latent space by labels. Our results show that a subset of latent dimensions consistently produced stronger visual changes than others for a given input image. However, we also observed that the set of most useful dimensions varied across different images. This reflects the fact that in a standard VAE, the latent space is not disentangled: the influence of each latent dimension depends on the local posterior around the specific input. In other words, the sensitivity ranking captures local importance rather than universal semantic factors.

3.4.3. OUT-OF-DISTRIBUTION PERFORMANCE

The performance of VAE on unseen data sets is summarized in Table 2. Compared to CIFAR-10, CIFAR-100 shows slightly higher error and divergence, while SVHN achieves lower MSE but also significantly reduced KL, illustrating that reconstruction error alone is not a reliable OOD signal.

Dataset	MSE	KL
CIFAR-10	0.0315	63.82
CIFAR-100	0.0332	65.46
SVHN	0.0130	48.41

Table 2. Reconstruction MSE and KL divergence of the VAE on in-distribution (CIFAR-10) and OOD datasets (CIFAR-100, SVHN).

We now evaluate the behavior of the GAN under different latent priors. Since GANs do not reconstruct given inputs, there is no gold standard for measuring reconstruction fidelity. Instead, we qualitatively visualize generated samples. As shown in Figure 6, the GAN performs well when sampling from the training prior $\mathcal{N}(0, I)$. However, when the variance of the prior is increased to $\mathcal{N}(0, 2I)$, the visual

quality of the outputs degrades noticeably. This highlights the GAN’s sensitivity to deviations from its training distribution and its limited generalization beyond the standard normal prior.

Table 3. VAE anomaly detection performance on CIFAR-10 (in-distribution) and SVHN (OOD) across different calibration percentiles.

Percentiles (lower, upper)	FPR (CIFAR-10)	TPR (SVHN)	AUROC	AUPRC
(0.00, 1.00)	0.0006	0.0004	0.255	0.749
(0.01, 0.99)	0.0198	0.0644	0.256	0.748
(0.02, 0.98)	0.0410	0.1300	0.252	0.747
(0.03, 0.97)	0.0626	0.1924	0.258	0.748
(0.04, 0.96)	0.0848	0.2507	0.256	0.748
(0.05, 0.95)	0.0964	0.2756	0.256	0.749
(0.06, 0.94)	0.1210	0.3204	0.258	0.749
(0.07, 0.93)	0.1456	0.3694	0.254	0.747
(0.08, 0.92)	0.1726	0.3987	0.258	0.749
(0.09, 0.91)	0.1888	0.4266	0.255	0.748
(0.10, 0.90)	0.2036	0.4383	0.252	0.747

As shown in Table 3, reconstruction-error based anomaly detection fails to reliably separate CIFAR-10 from SVHN. While increasing the quantile range improves the detection rate on SVHN, it also raises the false positive rate on CIFAR-10. Across all thresholds, AUROC remains close to random chance (≈ 0.25) and AUPRC hovers around 0.75, confirming that reconstruction error is not a robust OOD detection signal. Using the entire ELBO loss instead of simple mean squared error gives us a similar trend.

3.5. Contrastive Models (CLIP)

3.5.1. CIFAR-10 ZERO SHOT RESULTS

The pretrained CLIP ViT-B/32 model achieved an overall zero-shot accuracy of 88.77% on CIFAR-10. Performance was strongest on automobile (97.6%), horse (97.6%), ship (95.5%), and airplane (90.1%), categories with large and easily distinguishable shapes. In contrast, lower accuracies were observed for deer (76.5%) and frog (77.3%), reflecting difficulty with small animals in low-resolution images. Cat (86.0%) and dog (88.3%) also showed reduced accuracy due to overlapping features.

When compared with a fine-tuned ResNet, which reached 96.72%, the supervised model has the advantage of being directly optimized for CIFAR-10’s resolution and characteristics. However, CLIP’s near-90% accuracy without task-specific training highlights its strong zero-shot transfer ability and reliance on higher-level semantic features over fine-grained textures.

3.5.2. PACS ZERO-SHOT RESULTS

On the PACS dataset, CLIP demonstrates strong cross-domain generalization. Accuracy is nearly perfect on the Photo domain (99.82%), very high on Art Paintings (96.19%) and Cartoons (97.65%), and lower but still impressive on Sketches (84.91%). The decline in the sketch

domain is expected, as sketches remove texture and color, forcing reliance on global shape alone.

Overall, these results show that CLIP performs robustly across substantial domain shifts, maintaining high accuracy even when style and visual detail change dramatically. This highlights its ability to generalize beyond natural photographs, unlike conventional models that typically suffer under domain shift.

3.5.3. EVALUATION ACROSS PROMPT SETS

We then ran evaluations across all four prompt sets, each highlighting CLIP’s performance on CIFAR-10.

The different prompt styles for CLIP on CIFAR-10 show minor performance differences: the Simple Prompt achieves 87.41% accuracy, the Detailed Prompt 87.86%, the Photo Prompt performs best at 88.77%, and the Ensemble Prompt reaches 88.70%, indicating that more descriptive or combined prompts offer only slight gains over simple, photo-aligned phrasing.

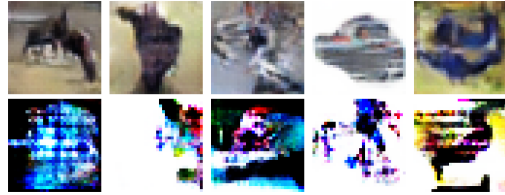


Figure 6. GAN samples generated from two latent priors: $\mathcal{N}(0, I)$ (top row) and $\mathcal{N}(0, 2I)$ (bottom row).

Overall, CLIP demonstrates robustness across all prompt sets, with performance consistently in the high 80s. The best results come from prompts aligned with natural image descriptions, but even minimal or overly detailed prompts maintain strong accuracy.

3.5.4. SKETCH RETRIEVAL RESULTS

CLIP demonstrated strong retrieval performance on the sketch setup. Out of seven text prompts, six were correctly matched to their corresponding images. This suggests some overlap in high-level line structures or biases in the training distribution.

For the image to text direction, results were even stronger: all sampled images were perfectly matched to their correct text descriptions. This highlights that while text to image retrieval can be more challenging due to prompt ambiguity and diverse visual candidates, image to text retrieval benefits from clearer visual cues and produced consistently accurate matches.

3.5.5. REPRESENTATION ANALYSIS RESULTS

t-SNE visualizations of CLIP embeddings show strong semantic organization across domains. While domain-specific groupings (e.g., sketches, cartoons, grayscale) are partly visible, images from the same category cluster together regardless of style. For example, sketches, photos, and grayscale renderings of the same object appear in close proximity, reflecting CLIP’s ability to encode category identity beyond low-level appearance. These results confirm that CLIP representations emphasize semantic coherence, consistent with its robust cross-domain retrieval performance.

3.5.6. TEXTURE ROBUSTNESS OF CLIP

CLIP demonstrated strong robustness to texture perturbations. When the global outline of the object was preserved, such as a cat overlaid with metallic textures, the model consistently predicted the correct class. However, performance degraded in cases where the applied textures visually distorted or overwhelmed the object’s shape. For example, applying bear fur blurred the silhouette, and bicycle spokes disrupted the natural contours, both of which led to misclassification. These results highlight CLIP’s reliance on shape and semantic cues, showing that its zero-shot predictions remain stable unless the overall structure is compromised.

3.5.7. ROBUSTNESS TO IMAGE PERTURBATIONS

Results showed that CLIP correctly predicted the image of a cat as a cat across every perturbation, including blurring, posterization, etc. This demonstrates that CLIP is robust to a wide variety of corruptions and distortions, maintaining stable predictions even when low-level texture, color, and fine details are heavily altered.

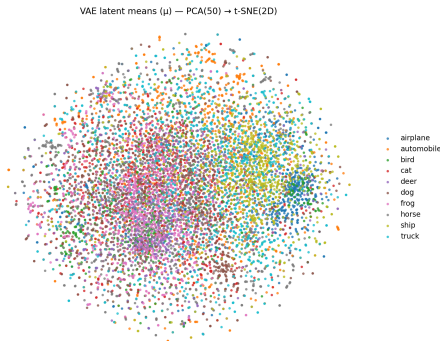


Figure 7. t-SNE visualization of latent mean vectors from the CIFAR-10 test set (best viewed electronically).

4. Discussion

4.1. CNN vs ViT Biases

Our study demonstrates that CNNs are semantically biased toward texture, while ViTs are more biased toward global shape and structure. This is evident from the better performance of ViT-B/16 on the cue-conflict dataset, which preserves shape information while removing texture. These observations align with prior work highlighting the strong inductive biases of CNNs, stemming from their local receptive fields and weight-sharing kernels (Wang & Wu, 2024). Furthermore, ResNet-50, pretrained on ImageNet texture bias of CNNs trained on this dataset, which is already well documented in literature. Thus, our findings are corroborated with extensive prior studies showing that ImageNet-trained CNNs rely heavily on texture cues due to high frequency information (Geirhos et al., 2022; Adaloglou & Kaiser, 2023). Evaluation on the grayscale dataset shows more robustness in the ViT, due to the spatial and shape structure being preserved. This is supported by literature, which suggests that ViTs are less affected by color changes due to their shape bias (Zhang et al., 2022). In summary, ViTs exhibit stronger shape bias whereas CNNs show stronger texture bias.

4.2. Architectural Biases

CNNs possess a strong inductive bias for translation invariance due to weight sharing, which is proved by the higher consistency on the shifted STL-10 dataset (Wang & Wu, 2024). On the other hand, ViTs do not have translation invariance because of their reliance on positional embeddings to provide spatial information (Zhang et al., 2022). Our experiments show the same trend with the lower consistency on the shifted dataset. Due to architectural biases, permutation sensitivity varies for models. The self-attention mechanism in ViTs allow it to model long-range correlations and global relationships and dynamic receptive fields, compared to the localized receptive fields on CNNs (Adaloglou & Kaiser, 2023). Thus, under patch shuffling and occlusion, ViTs show more robustness and higher accuracy across both modified datasets. CNNs, on the contrary, are not permutation invariant because shuffling/occluding pixels would remove any local structures that convolutions rely on (Wang & Wu, 2024).

4.3. Fidelity vs Diversity trade off

Our results highlight the fidelity diversity trade-off and this can be linked to the divergence each model optimizes. The VAE reconstructions (Fig. 3) retain the overall class structure and cover all CIFAR-10 categories, but are consistently blurry and have lost all sharp edges, as shown by the high FID score of 132. This can be attributed to the forward KL term in the ELBO. Forward KL penalises probability mass

that falls outside the prior heavily, but it does not strongly penalise missing individual modes in our learnt distribution. As a result, the posterior covers all the support of the latent space which ensures that all modes are represented. This prevents mode collapse and explains the diversity in VAE reconstructions. However, this same support-covering behaviour reduces sharpness, and when combined with the Gaussian likelihood, which averages across plausible outputs, it directly produces the smooth but blurry reconstructions we observe with no edges resulting in images looking like blobs.

GANs, in contrast, produces sharper images and achieves a much lower FID of 30, showing they are much closer to the truer distribution. But the class histogram of 10,000 generated samples (Fig. 4) shows imbalance, all classes appear, but some dominate. This partial mode collapse is due to the objective function the model optimises. As established in Mroueh & Sercu (2017), the original min-max GAN formulation is equivalent to minimising the Jensen-Shannon (JS) divergence. Li & Farnia (2023) classify JS as a uniformly mode-seeking divergence, in the same category as reverse KL. It is these mode seeking divergences that push the generator to produce high-quality samples from a subset of modes while ignoring others, explaining the sharpness of GAN outputs alongside their imbalanced coverage.

4.4. Training Dynamics

Our VAE training initially produced poor reconstructions because the KL term dominated the ELBO as we were giving it a weight of one which proved to be too strong of a KL penalty that forces the approximate posterior $q(z|x)$ to stay close to the prior $p(z)$, which weakens the information carried in the latent variable and leads to posterior collapse. As a result, the decoder cannot rely on meaningful latents and reconstruction quality degrades. This trade-off between reconstruction fidelity and latent regularization is explicitly demonstrated in the β -VAE framework. Higgins et al. (2017) show that scaling the KL term with a higher β encourages disentangled latent representations but at the expense of reconstruction quality. By applying KL annealing, where the KL weight is gradually ramped up from near zero to one, we allowed the model to first learn meaningful reconstructions under a stronger posterior before introducing the full regularisation term. This schedule improved reconstruction quality and yielded a latent space that could still be regularised for subsequent experiments.

For our GAN, we used the standard DCGAN loss with 50 epochs and a learning rate of 2×10^{-4} . Early experiments with batch normalization in the discriminator caused instability: the discriminator quickly overpowered the generator, destabilising the min-max game. We mitigated this by switching to spectral normalization in the discriminator.

As shown in Miyato et al. (2018), spectral norm prevents the discriminator from becoming too strong relative to the generator, stabilises training, and produces a healthier adversarial balance. Our improved results after making this change are consistent with the findings reported in that paper.

4.5. Latent Representation Discussion

In our experiments, both the VAE and the GAN produced smooth latent interpolations. For the VAE this behaviour is expected as the KL regularisation in the ELBO enforces that the approximate posterior stays close to a simple prior, typically $N(0, I)$. This motivates the latent space to be continuous and structured, so interpolations between latent codes almost always decode to valid samples. For the GAN, however, such smoothness is not guaranteed. As shown by Michelis & Becker (2021), the geometry of GAN latent spaces is highly non uniform. Some regions correspond to flatter, well-trained parts of the data distribution, while others are highly curved and less reliable. They show that the quality of interpolation depends strongly on where in latent space the operation is performed. Our result can therefore be explained by the fact that the two random latent codes we selected, and the linear path between them, likely lay in one of the flatter regions where the generator had sufficient coverage, producing smooth transitions. This is a local property, not a global guarantee, unlike the more structured latent space enforced by VAEs.

4.6. Zero-Shot Generalization and Domain Robustness

On CIFAR-10, CLIP shows strong overall performance despite never being trained on this dataset. Prompts framed in a photo-oriented style achieved the best results, highlighting that naturalistic phrasing aligns most closely with CLIP’s pretraining. This demonstrates that CLIP can map even low-resolution objects to semantic categories without elaborate prompt engineering. Vehicles and large animals were recognized most accurately, reflecting CLIP’s emphasis on global structure and shape cues. Smaller or visually overlapping animals were more challenging, as distinguishing them often depends on fine-grained textures that CLIP does not prioritize. Compared to a fine-tuned ResNet trained directly on CIFAR-10, CLIP fell short in absolute accuracy but demonstrated remarkable flexibility, transferring broad semantic knowledge zero-shot while ResNet relied on dataset-specific optimization. This contrast underscores CLIP’s human-like shape bias, which supports generalization beyond supervised CNNs.

Results on PACS reinforce these conclusions under explicit distribution shifts. CLIP achieved near-perfect accuracy on the Photo domain and Art Paintings and Cartoons, where stylization alters textures but preserves outlines. Its reliance on high-level structure and exposure to diverse imagery

during pretraining explain this robustness, further showing that multimodal pretraining aligns representations with semantic rather than purely pixel-level features. Sketches were the most challenging, stripping away almost all texture, shading, and color, yet CLIP maintained strong zero-shot accuracy above 84, vastly outperforming a ResNet-50 fine-tuned on PACS, which dropped to just over 53. This contrast highlights the models’ inductive biases: supervised CNNs remain texture-dependent, while CLIP generalizes via semantic abstraction.

Taken together, these findings show that contrastive multimodal pretraining shapes visual encoder biases toward shape and global structure, enabling robust transfer across tasks and domains. In direct relation to the fourth research question, CLIP demonstrates a human-like shape bias that underpins its ability to generalize zero-shot, even under severe distribution shifts where conventional supervised models struggle.

4.7. Texture and Perturbation Robustness

This analysis highlights an important aspect of CLIP’s behavior: it is significantly more shape-biased than traditional CNNs, yet its reliance on global semantics is not absolute. CLIP excels at recognizing large, distinct object outlines and mapping them to textual categories, allowing it to succeed when overall structure is preserved despite texture changes. When textures distort shapes or obscure semantic cues, predictions can shift. Bear fur overlays blur natural contours, and rigid bicycle textures interfere with smooth outlines, revealing that CLIP does not ignore texture but weights shape and semantics more heavily. Once shape cues are compromised, dominant textures can influence classification.

Compared to CNNs like ResNet, the difference is striking. On stylized images, ResNet achieves only 41.39% accuracy, reflecting its reliance on local textures, which limits generalization under domain shift. CLIP, trained on hundreds of millions of diverse image–text pairs, develops representations centered on global shape and semantic alignment. Exposure to sketches, cartoons, and varied contexts injects inductive bias through training data, not just architecture, enabling robust performance across multiple styles where ResNet fails.

Perturbation experiments reinforce this robustness: blurred, posterized, solarized, grayscale, and edge-only images are consistently classified correctly, demonstrating that CLIP relies on high-level semantic features rather than fine textures or colors.

4.8. Representation Quality of CLIP vs. ResNet Embeddings

The t-SNE visualizations reveal key differences in how CLIP and ResNet organize their feature spaces, with implications for robustness and generalization across domains.

For CLIP, embeddings colored by domain show local clustering of images from the same domain (e.g., sketches, cartoons, grayscale) due to shared low-level statistics. More strikingly, when colored by semantic class, images from the same category (e.g., cats or airplanes) cluster together across domains, reflecting CLIP’s ability to abstract away from stylistic variation and encode high-level conceptual identity. This aligns with retrieval experiments, where CLIP successfully matched sketches to their correct text descriptions, confirming that its representation space prioritizes semantics over appearance.

In contrast, ResNet-50 t-SNE plots show that while natural in-domain images form clear class-wise clusters, introducing stylized variants disrupts this structure. Some stylized samples remain partially aligned with natural counterparts, but most collapse toward a central region, losing class distinctions. Stylized dogs, airplanes, and cars are projected into an undifferentiated space rather than forming meaningful clusters. This illustrates ResNet’s brittleness under distribution shifts: its texture bias causes it to discard shape- and semantics-based cues, failing to preserve categorical separation.

Overall, CLIP’s embeddings emphasize object shape, global structure, and conceptual identity, bridging stylistic gaps. ResNet, trained on CIFAR-10, inherits a narrow domain-specific bias, making its features fragile under altered appearances.

This comparison shows that CLIP’s feature space is not only class-discriminative but also semantically structured, supporting generalization to unseen domains. ResNet maintains in-domain clustering but collapses under distribution shifts, whereas CLIP’s abstract and transferable representations explain its robustness to sketches, cartoons, and other stylized inputs.

4.9. Generalization

For discriminative models, CNNs and ViTs, ViTs better generalize to OOD data as evident by its minimal accuracy shift on domain shifted data like the stylized dataset. This is supported by previous literature, which suggests that shape-biased models like ViTs are more robust to distribution shifts compared to CNNs (Geirhos et al., 2022; Alijani et al., 2024). Moreover, ViTs show higher robustness to permutations as mentioned above, which also contribute to their better performance under distribution shifts. Therefore, CNN model fails on the sketch set, which is supported by lit-

erature suggesting the these models fail to recognize shapes were texture cues are missing (Geirhos et al., 2022). However, ViTs have weaker inductive biases, which make them data hungry. This can be linked to the failure of the ViT on the sketch domain, which at a naive glance was unexpected as it contradicted previous literature and do not show the ViT’s reliance on shape. The ViT was expected to perform better on the sketch domain the ResNet since the sketch domain only contained outlines/edges, which the model would rely on. However, it performed even worse, which could be attributed to its weak inductive bias requiring large amounts of data and sub-optimal training regime. This was further supported by literature that showed that texture bias in CNNs is not inherent but rather learned from datasets like ImageNet (Geirhos et al., 2022). Thus, it highlights the biases are not necessarily absolute and are not guaranteed to show up the same way everywhere, highlighting the importance of training and training data on the generalization of models.

Moreover, When we evaluated our VAE on CIFAR-100, the reconstructions appeared as distorted versions of CIFAR-10 classes. This is expected, the model has only learned CIFAR-10 and forces unfamiliar inputs into that space. The performance drop was modest because CIFAR-100 shares similar low-level statistics with CIFAR-10, both are natural images with comparable means and variances, so the model can still encode and decode them somewhat accurately. On SVHN, however, the story was completely different. Reconstructions looked digit-like and achieved lower error than CIFAR-10 itself. This counter-intuitive result is exactly what Nalisnick et al. (2019) report, generative models trained on CIFAR-10 assign higher likelihood to SVHN because likelihoods are dominated by low-level statistical properties rather than semantic similarity. SVHN has similar mean pixel values but much lower variance, making it statistically simpler and therefore easier for the model to explain under its Gaussian likelihood. This shows that reconstruction error is not a reliable OOD signal: some OOD inputs (CIFAR-100) are reconstructed poorly, while others (SVHN) appear easier than the training data itself. Robust anomaly detection requires setting thresholds on both ends of the error distribution and calibrating against multiple OOD datasets, not relying on a single benchmark.

These results demonstrate that our VAE generalises better to datasets that share similar or simpler statistical structures compared to the training set. When the pixel-level statistics of the new dataset fall within or inside the variability of CIFAR-10, the model can encode and decode them with relatively low error, even if the semantic content is completely different. This further confirms that likelihood-based VAEs are primarily sensitive to low-level statistics rather than high-level semantics. However as we have not really tested on a data set that brings much harder statistics than CIFAR-10 we cannot be certain that our hypothesis holds.

CLIP’s generalization provides a clear example of how human-aligned inductive biases affect out-of-distribution (OOD) performance. Unlike CNNs that rely heavily on local textures and fine-grained pixel patterns, CLIP emphasizes global shape, object outlines, and high-level semantic cues. This shape- and concept-oriented bias allows the model to correctly classify objects even when low-level visual features are altered, making it robust to stylization, blurring, grayscale conversion, and other texture perturbations. For instance, images of cats or vehicles maintain accurate classification despite heavy modifications to texture or color, demonstrating that CLIP prioritizes the overall structure and conceptual identity of objects over superficial appearance.

Across datasets like CIFAR-10, PACS, and various stylized images, including sketches, cartoons, and paintings, CLIP consistently outperforms supervised CNNs such as ResNet, especially under OOD conditions. While ResNet embeddings collapse for stylized inputs, CLIP embeddings cluster by semantic class rather than visual domain, indicating that its representation space encodes abstract, transferable concepts. This enables zero-shot generalization to entirely unseen domains, confirming that human-like inductive biases, here, a focus on global features and semantics, enhance OOD robustness.

Taken together, CLIP illustrates a key trade-off in inductive bias design: by learning human-like, shape- and semantics-focused representations, the model achieves strong OOD generalization at the cost of some sensitivity to fine-grained details. Its large-scale image–text pretraining provides the diverse exposure necessary to develop these biases, confirming that both architectural design and training distribution jointly determine the extent of OOD robustness.

Table 4. Zero-Shot CLIP Accuracy on CIFAR-10 Test Set

Class	Accuracy (%)	Correct / Total
Airplane	90.10	901 / 1000
Automobile	97.60	976 / 1000
Bird	88.90	889 / 1000
Cat	86.00	860 / 1000
Deer	76.50	765 / 1000
Dog	88.30	883 / 1000
Frog	77.30	773 / 1000
Horse	97.60	976 / 1000
Ship	95.50	955 / 1000
Truck	89.90	899 / 1000
Overall	88.77	8877 / 10000

5. Conclusion

Model architecture and training data significantly impact generalization. Understanding the inductive biases of different models is important for designing robust systems that

perform well under domain shifts. We show CNNs have strong inductive biases towards texture and locality, and remain translation invariant because of weight-sharing convolutions. On the other hand, ViTs, are more shape-based because of global attention and less sensitive to permutations, but their weaker inductive biases make them data hungry. However, because of this and their human-like shape bias they can perform better on OOD data.

Similarly, contrastive models like CLIP show how large-scale image–text pretraining shapes strong biases: CLIP prioritizes global shape and semantics, enabling robust zero-shot generalization across datasets and stylized domains, unlike CNNs that rely on texture. Generative models reveal another dimension. VAEs, via forward KL, are mode-covering—capturing diversity but producing blurry reconstructions—suited for coverage and anomaly detection. GANs, by contrast, are mode-seeking, generating sharp images but risking mode collapse. These divergence-driven biases shape OOD behavior, with VAEs over-scoring simpler datasets (e.g., SVHN) and GANs ignoring modes, highlighting the fidelity–diversity trade-off in generalization.

These findings suggest that carefully incorporating inductive biases into model design and training can improve robustness. Moving forward, we can design hybrid architectures that incorporate strengths from different models.

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